

Homework 3

Alex Ojemann

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Problem 1

Part a

Roots of $\phi(x)$: $x = 10i/9, x = -10i/9$

Roots of $\theta(x)$: $x = -3$

No redundancy, this is ARMA(2,1)

This is causal and invertible because all roots of $\phi(x)$ and $\theta(x)$ are greater than one in magnitude.

Part b

Roots of $\phi(x)$: $x = 1$

Roots of $\theta(x)$: $x = 1, x = -2$

The root $x=1$ is redundant so the final model is MA(1) ($X_t = W_t + W_{t-1}/2$)

This is causal because all MA processes are causal. This invertible because all roots of $\theta(x)$ are greater than one in magnitude.

Part c

Roots of $\phi(x)$: $x = 1/3$

Roots of $\theta(x)$: $x = 1/2, -1/4$

No redundancy, this is ARMA(1,2)

This is not causal nor invertible because some roots of $\phi(x)$ and $\theta(x)$ are less than or equal to one in magnitude.

Part d

Roots of $\phi(x)$: $x = 1/2 + \sqrt{3}i/2, 1/2 - \sqrt{3}i/2$

Roots of $\theta(x)$: $x = 9/8$

No redundancy, this is ARMA(2,1)

This invertible but not causal because all roots of $\theta(x)$ but not $\phi(x)$ are greater than one in magnitude.

Part e

Roots of $\phi(x)$: $x = 1/2, x = -1/2$

Roots of $\theta(x)$: $x = 1 + i, 1 - i$

No redundancy, this is ARMA(2,2)

This invertible but not causal because all roots of $\theta(x)$ but not $\phi(x)$ are greater than one in magnitude.

Part f

Roots of $\phi(x)$: $x = 4/3, x = -1/3$

Roots of $\theta(x)$: N/A

No redundancy, this is AR(2)

This invertible because all AR processes are invertible. This is not causal because not all roots of $\phi(x)$ are greater than one in magnitude.

Problem 2

Only the models from part a and b are causal.

Model from part a

$$\psi_0 = 1$$

$$\psi_1 = 1/3$$

$$\psi_2 = -0.81\psi_0 = 0.81$$

$$\psi_3 = -0.81\psi_1 = -0.27$$

$$\psi_4 = -0.81\psi_2 = 0.6561$$

Model from part b

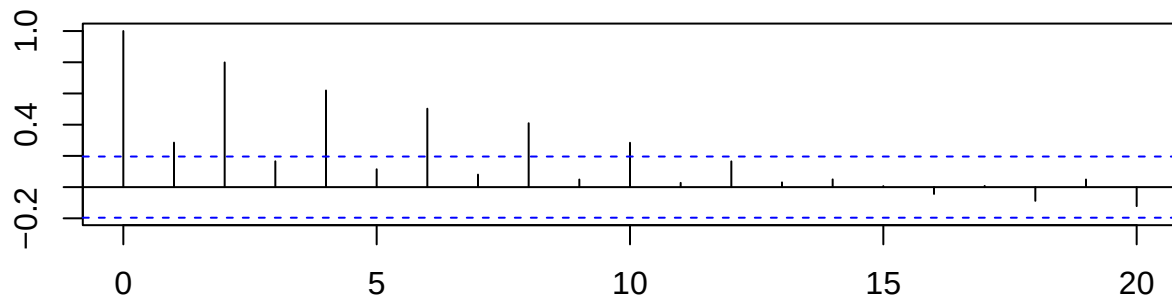
$$\psi_0 = 1$$

$$\psi_1 = 1/2$$

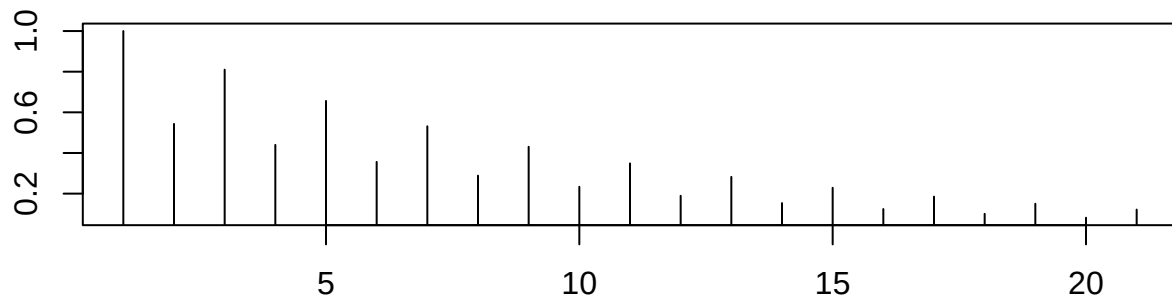
$$\psi_2 = \psi_3 = \psi_4 = 0 \text{ because this is an MA(1) model.}$$

Problem 3

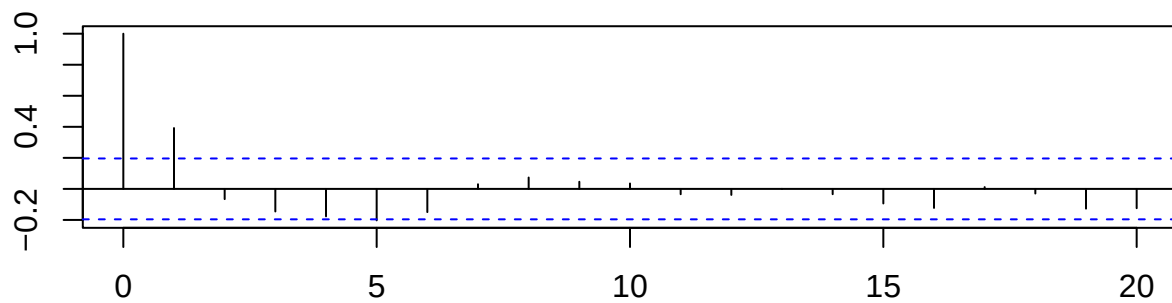
Sample ACF for ARMA(2,1) (Part a)



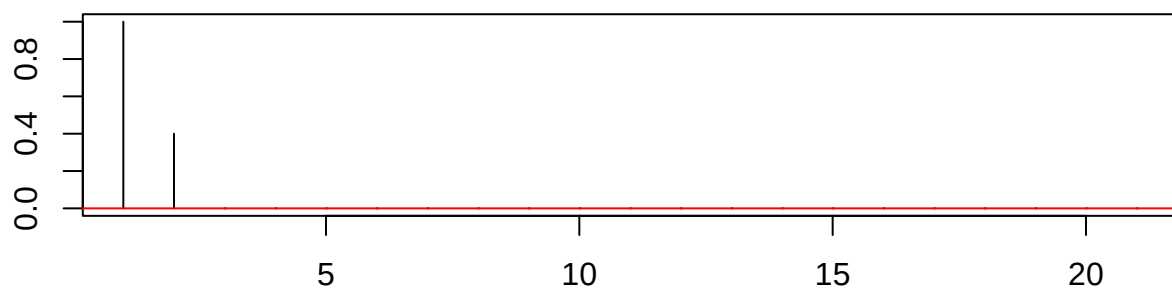
Theoretical ACF for ARMA(2,1) (Part a)



Sample ACF for MA(1) (Part b)



Theoretical ACF for MA(1) (Part b)



Problem 4

Part a

We know that $1 - \phi_1 z - \phi_2 z^2 = (1 - (1/z_1)z)(1 - (1/z_2)z)$

Based on this, $\phi_1 = 2 \cos(\theta_z)/|z_1|$, $\phi_2 = 1/|z_1|^2$

$|z_1|$ determines the rate of decay of the ACF. We will set that to 4 to ensure that the ACF decays quickly enough for the series to be stationary. When smaller values of $|z_1|$ were chosen, the arima.sim method in the next part throws an error because the series is not stationary.

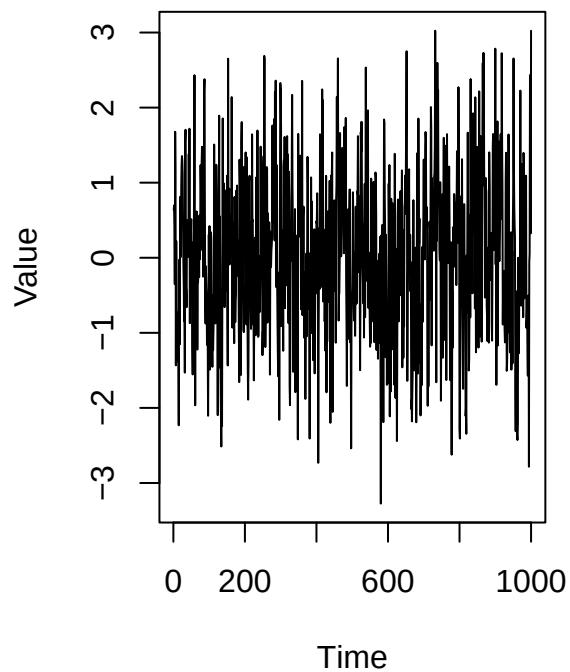
$\theta_z = 2\pi/P = 2\pi/9$ (P is the period).

$\phi_1 = 2 \cos(2\pi/9)/|4| = 0.383$

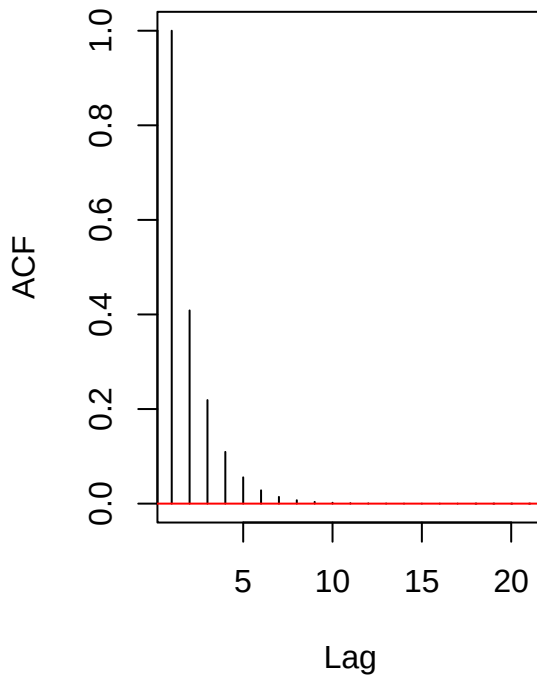
$\phi_2 = 1/|4|^2 = 0.0625$

Parts b and c

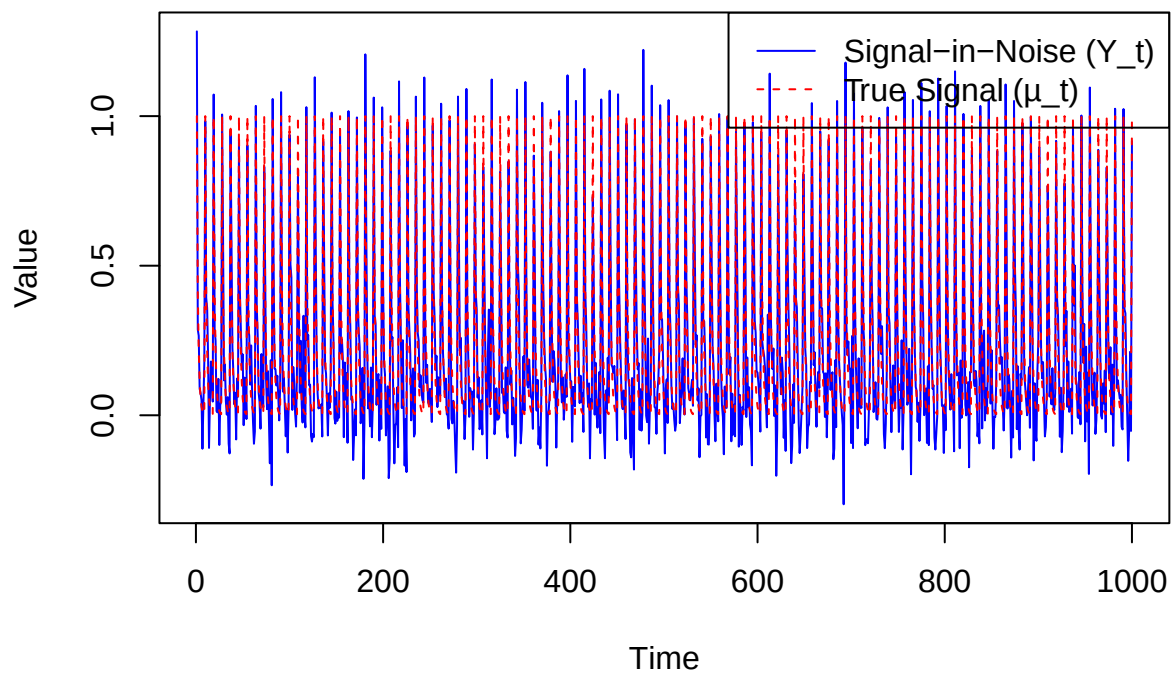
Simulated AR(2) Time Series



Theoretical ACF for AR(2)



Signal-in-Noise Time Series



Part d

It is extremely difficult to tell where the peaks and valleys are in the ACF after the first one because the rate of decay is relatively large. As mentioned previously, this large decay rate was chosen because the `arma.sim` method throws an error because the series is not stationary for smaller decay rates.

However, we can see the period much more clearly in the signal-in-noise plot, where there appears to be a maximum every nine iterations as expected.